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Subject: Manuscript submission to Bioinformatics

Dear Editor,

I kindly ask that you consider our manuscript, "*GaugeFixer: overcoming parameter non-identifiability in models of sequence-function relationships,*" for publication as an Original Paper in Bioinformatics.

Mathematical models that describe sequence-function relationships are widely used in computational biology, from predicting transcription factor binding sites to modeling the effects of genetic variation on protein function. A fundamental challenge when interpreting these models is that their parameters are not uniquely determined by data. Before parameter values can be meaningfully interpreted or compared across models, these ambiguities must be resolved through a procedure called “gauge-fixing”. Despite the importance of this issue, no general-purpose computational tools for gauge fixing have been available.

My lab recently developed the first general mathematical theory of gauge-fixing for a large class of commonly used sequence-to-function models (Posfai et al., 2025, PLoS Comp Bio, PMID: 38798671). The direct computational implementation of these methods, however, is often infeasible due to the necessary matrix operations having memory and runtime requirements that scale quadratically with the number of parameters.

Our manuscript introduces GaugeFixer, an open-source Python package that addresses this need. GaugeFixer implements a computationally efficient algorithm that exploits the mathematical structure of this theory to achieve linear scaling in both memory and computation time. GaugeFixer thus enables rapid gauge fixing for models with millions of parameters—a task that would otherwise be computationally infeasible. We demonstrate the biological utility of GaugeFixer by interpreting a complex fitness landscape (~2M parameters) for 5’ UTRs, revealing fine-scale variation in ribosome binding preferences across registers. GaugeFixer thus enables biological interpretations of genotype-phenotype maps that were previously intractable.

My lab has a track record of building and maintaining robust and highly cited software, including Logomaker (Tareen and Kinney, Bioinformatics, 2020; **501 citations**). We have similarly taken care to make GaugeFixer robust, user friendly, open source, easily installable, and well documented.

We suggest the following experts as potential reviewers:
- Donate Weghorn, CRG Barcelona, donate.weghorn@crg.eu
- Michael Desai, Harvard University, mdesai@oeb.harvard.edu
- Anne-Florence Bitbol, EPFL, anne-florence.bitbol@epfl.ch
- Harmen Bussemaker, Columbia University, hjb2004@columbia.edu
- Armita Nourmohammad, University of Washington, armita@uw.edu

My coauthors and I look forward to hearing back from you.

Sincerely,

Justin Block Kinney

Justin B. Kinney, PhD